

Sex differences in hemodynamic response to high-intensity interval exercise

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Abstract

Sex differences in the cardiorespiratory and hemodynamic response to exercise exist due to differences in heart size, blood volume, and hemoglobin mass, eliciting higher maximal oxygen uptake ($\text{VO}_{2\text{max}}$) in men versus women. Data are equivocal whether sex differences in training responsiveness occur. This study investigated potential sex differences in the hemodynamic response (stroke volume (SV) and cardiac output (CO)) to high-intensity interval exercise (HIIE). Habitually active men ($n=15$) and women ($n=13$) underwent $\text{VO}_{2\text{max}}$ testing, followed by three HIIE sessions consisting of the 4×4 , 10×1 , and reduced exertion high-intensity training (REHIT), whose order was randomized. During exercise, oxygen uptake (VO_2) and hemodynamic responses were determined. Results showed no sex difference in peak relative VO_2 ($p=0.263$), CO ($p=0.277$), or SV ($p=0.116$), although absolute values were higher in men ($p<0.05$). Peak absolute (127.3 ± 20.6 vs. 115.2 ± 16.6 mL/beat, $p=0.004$, $d=0.66$) and relative SV (111.0 ± 15.5 vs. $100.7\pm 11.1\%$ max, $p=0.005$, $d=0.78$) were higher with REHIT versus 4×4 . No sex differences in mean relative VO_2 , CO, or SV occurred ($p>0.05$). Data showed lower mean VO_2 during REHIT versus 4×4 (59.3 ± 6.8 vs. 65.8 ± 5.8 % $\text{VO}_{2\text{max}}$, $p<0.001$, $d=1.05$) and 10×1 (59.3 ± 6.8 vs. 69.1 ± 7.4 % $\text{VO}_{2\text{max}}$, $p<0.001$, $d=1.4$). Mean CO was lower in REHIT than 10×1 (79.8 ± 8.6 vs. $84.0\pm 7.4\%$ max, $p=0.012$, $d=0.53$). Previously reported differences in $\text{VO}_{2\text{max}}$ response to HIIE may not be due to unique hemodynamic responses.

KEYWORDS

cardiac output, high-intensity interval exercise, intermittent exercise, oxygen uptake, sprint interval training, stroke volume

1 | INTRODUCTION

Compared to women, men have a greater heart size, blood volume, and hemoglobin mass¹ that elicits a 10% higher maximal oxygen uptake ($\text{VO}_{2\text{max}}$).² Prior data show a significantly greater increase in $\text{VO}_{2\text{max}}$ in men compared to women (22% vs. 15%) after 1 year of moderate-intensity

continuous training (MICT). Diaz-Canestro and Montero³ reviewed eight studies describing the $\text{VO}_{2\text{max}}$ response to chronic MICT. These authors concluded that MICT confers superior increases in $\text{VO}_{2\text{max}}$ in men versus women, although in three studies, women exhibited greater increases in $\text{VO}_{2\text{max}}$ versus men.^{4–6} These unique responses to training are plausible considering the aforementioned

differences in the capacity for O₂ transport between sexes.^{1,7,8} For example, women tend to have smaller left ventricular mass (25%–38%) in comparison to men, which is consistent with smaller dimensions and wall thickness.⁹ This difference may be due to discrepancies in body size, hormone levels, and hypertrophy of myocytes.^{7,9} Overall, these alterations in cardiac structure and function are important in mediating differences in VO_{2max}, as the Fick Equation ($VO_2 = (HR \times SV) \times a-vO_2$, where cardiac output (CO) = $HR \times SV$) denotes that oxygen uptake is determined by CO and peripheral oxygen extraction.

One primary barrier to MICT is a perceived lack of time.¹⁰ Only 47% of adults meet the aerobic physical activity guidelines, and 24% meet both aerobic and resistance training guidelines.¹¹ A time-saving alternative to MICT is high-intensity interval training (HIIT) which elicits a comparable or superior increase in VO_{2max} with a lower training volume versus MICT.¹² HIIT requires short (10s–5 min) bouts of vigorous efforts at intensities slightly below, at, or greater than those associated with VO_{2max} interspersed with recovery.¹³ The increase in VO_{2max} resulting from MICT is due to an increase in SV and CO and not a-vO₂diff.¹⁴ Similarly, recent data from a systematic review and meta-analysis exhibit that significant increases in VO_{2max} in response to HIIT are also associated with changes in SV and CO,¹⁵ suggesting that a greater oxygen delivery mediates the increase in cardiorespiratory fitness resulting from interval-based exercise.

Differences in oxidative capacity and fatigue resistance also occur between men and women. For example, Hottenrott et al.¹⁶ showed lower blood lactate concentration (BLa) and greater fatigue resistance in women completing repeated Wingate tests versus men, likely related to a reduced reliance on nonoxidative metabolism during exercise versus men. These innate physiological differences may elicit potential sex differences in the adaptive response to HIIT, as the acute response to exercise may mediate part of the chronic response.¹⁷ For example, 12 weeks of sprint interval training (SIT) led to attenuated increases in VO_{2max} in men versus women (6% vs. 19%),¹⁸ which is opposed by recent data exhibiting a greater increase in VO_{2max} in men compared to women (28% vs. 14%) undergoing 12 weeks of low-volume SIT.¹⁹ However, another study reported no sex difference in VO_{2max} response to six sessions of SIT in men and women with similar baseline VO_{2max}.²⁰ Notably, these studies are characterized by small sample sizes, which may limit the ability to detect potential sex differences in the training response.

The acute changes in CO and stroke volume (SV) are related to chronic cardiovascular responses to exercise training.²¹ Cardiac output increases markedly from rest (5 L/min) to approximately 20–40 L/min at VO_{2max}, which contributes to the variability in VO_{2max} across

participants.²² Similarly, SV increases during exercise but typically plateaus at approximately 50% VO_{2max},²¹ and the continued rise in HR mediates the subsequent increase in CO. Prior studies^{23,24} show that acute bouts of HIIE (high-intensity interval exercise) elicit significantly higher and near-maximal CO compared to continuous exercise. Overall, this substantial stress on the cardiovascular system is important for optimizing increases in VO_{2max} with interval training.²⁵

To our knowledge, no study has documented potential sex differences in the CO response to acute HIIE which may shed light on the limiting factors to training-induced changes in VO_{2max}. Since HIIT leads to significant increases in VO_{2max} in many adults,^{12,13,15,19} its adoption may have profound effects on population health. However, if men and women reveal discrepant cardiovascular responses to acute HIIE, this may partially mediate the previously reported sex differences in VO_{2max} in response to HIIT.^{18,19} It is possible that exercise prescription may need to elicit higher relative workloads for women to elicit similar hemodynamic responses and in turn, similar increases in VO_{2max} compared to men.

The primary aim of this study was to examine potential sex differences in the acute CO response to HIIE, as CO is the primary determinant of VO_{2max}.²⁶ The secondary aim was to identify differences in acute hemodynamic responses between unique HIIE protocols. It was hypothesized that a higher CO and SV (expressed as %CO_{max} and %SV_{max}) will occur in women versus men in response to interval exercise, due to their smaller heart size and lower blood volume which may elicit a greater relative load on the heart compared to men. A secondary hypothesis is that there will be a lower CO_{max} in response to low-volume sprint interval exercise (SIE) than in response to high volume HIIE protocols, as shown in a prior study.²⁴

2 | MATERIALS AND METHODS

2.1 | Participants

Initially, 33 men ($n=17$) and women ($n=16$) provided informed consent and participated in VO_{2max} testing, but five withdrew from the study for personal reasons. Fifteen men and 13 women subsequently completed all requirements of this study. Their physical characteristics are revealed in Table 1. They were 20–49 years old and were not obese (percent body fat $\leq 25\%$ in men and $\leq 33\%$ in women).²⁷ Each participant performed 150 min/week of moderate physical activity or 75 min/week of vigorous physical activity in the previous year, which was verified with a questionnaire. They completed resistance training, aerobic exercise, non-competitive sport, and/or surfing,

TABLE 1 Participant ($n = 28$) physical characteristics (mean $\pm$ SD).

Parameter	Men ($n = 15$)	Women ($n = 13$)	Range
Age (yr)	29 $\pm$ 8*	22 $\pm$ 2	20–49
Physical activity (h/week)	6 $\pm$ 3	6 $\pm$ 3	1.5–12.0
Mass (kg)	78.9 $\pm$ 9.8*	60.1 $\pm$ 7.3	47.0–94.8
Body mass index (kg/m ²)	25.0 $\pm$ 2.6*	21.9 $\pm$ 2.5	18.3–31.0
Body fat (%)	14 $\pm$ 4*	22 $\pm$ 4	7–26
VO _{2max} (L/min)	3.1 $\pm$ 0.5*	2.3 $\pm$ 0.3	1.86–3.97
VO _{2max} (mL/kg/min)	39.5 $\pm$ 3.4	38.2 $\pm$ 4.6	29.9–46.3
PPO (W)	293 $\pm$ 42*	217 $\pm$ 25	188–341
PPO (W/kg)	3.7 $\pm$ 0.3	3.6 $\pm$ 0.5	2.9–4.3

* $p < 0.05$ versus women.

yet none trained for a particular sport. They were also healthy and free of any condition which may preclude intense cycling. Participants provided written informed consent and filled out a health history questionnaire and PARQ to confirm eligibility. This protocol was approved by the University Institutional Review Board.

2.2 | Experimental design

This study used a randomized, within-subjects crossover design. On day one, VO_{2max} was determined during ramp exercise and a subsequent verification (VER) test. HIIE protocols varying in volume and structure were performed on three subsequent days at the same time within participants. During all sessions, pulmonary gas exchange data were acquired, and thoracic impedance was used to assess the hemodynamic response to exercise. Before all sessions, participants refrained from strenuous exercise for 24h, fasted for 2h, and were instructed to consume 1 L of water in the 2h prior. Participants also recorded all food and drink consumption 24h before the initial HIIE session and were encouraged to replicate this intake before the last two sessions. All sessions were separated by at least 48 h.

2.3 | Assessment of VO_{2max} and body composition

Initially, height and weight were assessed to determine body mass index (BMI in kg/m²). Percent body fat was calculated by measuring subcutaneous fat at three sites in rotational order, chest, abdomen and thigh in men and, triceps, supra-iliac and thigh in women.²⁸ Body density and percent body fat were calculated using sex and

race-specific equations. Before exercise, the metabolic cart (ParvoMedics True One, Sandy, UT) was calibrated according to manufacturer guidelines. During exercise, VO₂, carbon dioxide output (VCO₂), ventilation, and RER were measured every 10 s.

A ramp-based test was conducted on an electrically-braked cycle ergometer (Velotron Dynafit Pro, Quarq, Spearfish, SD) during which pulmonary gas exchange data were obtained using the metabolic cart. Participants began with a 2-min warm-up at 40–70 W and work rate was increased by 20–35 W/min until cadence declined below 50 rev/min. Peak power output (PPO) and maximal HR were identified as the peak values attained when cadence declined below 50 rev/min. Participants underwent a cooldown for 2 min at 50 W followed by a 5 min passive recovery. Participants then pedaled for 2 min at 20% PPO followed by 1 min at 50% PPO. Intensity was immediately increased to 85% PPO and participants cycled until volitional fatigue in order to verify VO_{2max} attainment.²⁹ Volitional fatigue was determined when cadence declined below 50 rev/min. VO_{2max} was identified as the mean of the ramp and VER-derived values, with maximum values from the ramp and VER tests determined as the average of the three highest consecutive values at volitional exhaustion. BLA was measured 3 min after the VER test from a 0.7 μ L blood sample via a fingertip puncture using a lancet (Owen Mumford Inc.) and portable analyzer (Lactate Plus, Sports Research Group, New Rochelle, NY). Prior to measurement, the fingertip was cleaned with a damp paper towel, then it was dried, punctured, and the first drop of blood was wiped away.

2.4 | Assessment of thoracic impedance

An impedance cardiograph device (PhysioFlow Enduro, Manatec) was used to estimate CO during all sessions.^{30,31} Prior to exercise, participants sat quietly for 5 min after which resting blood pressure (Omron Tru-Gage Cuff, Omron Healthcare) was determined twice at the brachial artery, with a 1 min period between measurements. Then, a total of six electrodes (PhysioFlow Versa Trode, Nissha Medical Technologies) were placed on each participant, two on the left supraclavicular fossa at the base of the neck, one on the right pectoralis major, one on the left rib closest to V6, and two directly left of the spine (on the transverse xiphoid process line and directly below). Each site was cleaned with an alcohol swab then an electrode gel (NuPrep; Weaver and Company) was rubbed into the skin. A 30-beat calibration procedure ensued to estimate resting HR, SV, and CO. During exercise, HR, CO, and SV were averaged every 10 s. Intraclass correlations for repeated measures of SV and CO at VO_{2max} in our lab

from similarly-trained men and women performing ramp exercise on the identical cycle ergometer are equal to 0.90–0.95.³¹ In addition, the minimum difference in CO in response to Wingate tests performed on separate days is equal to 1.8 L/min.³¹ Similar to previous studies,²⁴ hemodynamic variables and VO_2 were reported as mean and peak values and expressed as a percent of the maximal value derived from $\text{VO}_{2\text{max}}$ testing, which are considered the relative values. Peak values were determined as the highest of the average of three consecutive 10-s values at either 25%, 50%, 75%, or 100% of protocol duration. Mean values were determined as the average of data from the entire session, not including pre-exercise, warmup, or cool-down. Arteriovenous oxygen difference ($\text{a-vO}_2\text{diff}$) was calculated as the quotient of $\text{VO}_{2\text{max}}$ (mL min^{-1})/CO (L min^{-1}) and expressed in mL/dL.

2.5 | Description of HIIE protocols

All protocols were performed at the same time of day within participants (0600 a.m.–1700 p.m.) and completed in a randomized order (Table 2). Each protocol included a 3 min warm-up at 20% PPO and a 2 min recovery post-exercise. BL_a was measured pre-exercise after a 5 min seated rest, during exercise halfway through the protocol, and 3 min post-exercise using the same methods as denoted previously. During the 10 × 1 protocol,³² participants performed ten 1 min bouts of cycling at 85% PPO followed by 1 min of recovery at 20% PPO. The 4 × 4 protocol^{33,34} consisted of four 4 min bouts of cycling at 85%–95% HR_{max} separated by 3 min at 20% PPO. Power output was continuously altered during each interval to keep HR within this range, with absolute power output being reduced from bout 1 (~80%–90% PPO) to 4 (~60% PPO).³⁴ The third session consisted of reduced exertion high-intensity training (REHIT)³⁵ requiring two 20-s all-out sprints at workload = 5% body mass within a 10-min session.

2.6 | Statistical analysis

Data are expressed as mean ± SD and were analyzed using SPSS Version 28.0 (IBM). Normality was assessed using

the Shapiro–Wilk test. An unpaired *t*-test was used to compare demographic and maximal gas exchange data between men and women. Two-way repeated measures ANOVA was used to compare differences in the SV and CO response between men and women, with one within-subjects factor equal to HIIE condition (3 levels), and the between-subjects factor equal to sex (men vs. women). Similar analyses were conducted for the peak and mean value of these outcomes. If a significant F ratio was obtained, Tukey's post hoc test was used to detect differences between means. Cohen's *d* was used as an estimate of effect size between 0.15 and <0.40 representing “small,” between 0.40 and <0.75 representing “moderate,” and ≥0.75 representing “large.” Our sample size per group is similar to or larger than prior studies assessing sex differences in physiological responses³⁶ or examining changes in SV and CO.^{19,20,24,31} Statistical significance was set at $p < 0.05$.

3 | RESULTS

3.1 | Maximal gas exchange and hemodynamic data

Table 3 shows that men exhibited higher ($p < 0.05$) CO_{max} , SV_{max} , and $\text{a-vO}_2\text{diff}_{\text{max}}$ than women, although HR_{max} , cardiac index (CI_{max}), and BL_a were similar ($p > 0.05$).

3.2 | Peak hemodynamic and metabolic data

Figure 1 reveals aggregate changes in HR, SV, CO, and VO_2 in response to various HIIE protocols in our participants. There was no main effect of protocol ($p = 0.197$) or protocol X sex interaction ($p = 0.469$) for peak VO_2 (L/min), yet a main effect of sex occurred as men showed greater peak VO_2 than women ($p < 0.001$) (Table 4). No main effect of protocol ($p = 0.478$), sex ($p = 0.263$), or protocol X sex interaction ($p = 0.744$) occurred for relative VO_2 ($\%\text{VO}_{2\text{max}}$). Peak absolute CO was higher in men versus women ($p = 0.012$), but there was no effect of protocol ($p = 0.094$) or protocol X sex interaction ($p = 0.445$). Relative CO ($\%\text{CO}_{\text{max}}$) showed no main effect of protocol

Protocol	Bout duration (seconds)	Recovery duration (seconds)	Intensity (%PPO, % HR_{max} , or %BM)	Total duration (minutes)
10 × 1	60	60	85%PPO	23
4 × 4	240	180	85%–95% HR_{max}	28
REHIT	20	180	5%BM	10

Abbreviations: BM, body mass; PPO, peak power output; HR, heart rate.

TABLE 2 Summary of the HIIE protocols used in this study.

TABLE 3 Maximal gas exchange, hemodynamic, and BLA data between men and women (mean $\pm$ SD).

Variable	Men (<i>n</i> = 15)	Women (<i>n</i> = 13)	<i>p</i> Value	Cohen's <i>d</i>
HR _{max} (b/min)	178 $\pm$ 14	184 $\pm$ 9	0.07	0.59
CO _{max} (L/min)	21.6 $\pm$ 3.6*	18.8 $\pm$ 3.2	0.02	0.86
SV _{max} (mL/beat)	126.6 $\pm$ 20.2*	105.8 $\pm$ 17.7	0.004	1.14
CI _{max} (L/min/m ²)	11.0 $\pm$ 1.4	11.3 $\pm$ 1.7	0.59	0.21
a-vO ₂ diff max (mL/dL)	14.6 $\pm$ 2.4*	12.3 $\pm$ 1.5	0.006	1.17
BLA (mmol/L)	11.9 $\pm$ 2.4	10.3 $\pm$ 1.8	0.06	0.77

Abbreviations: a-vO₂diff, arteriovenous oxygen difference; BLA, blood lactate concentration; CO, cardiac output; HR, heart rate; PPO, peak power output; SV, stroke volume.

**p* < 0.05 versus women.

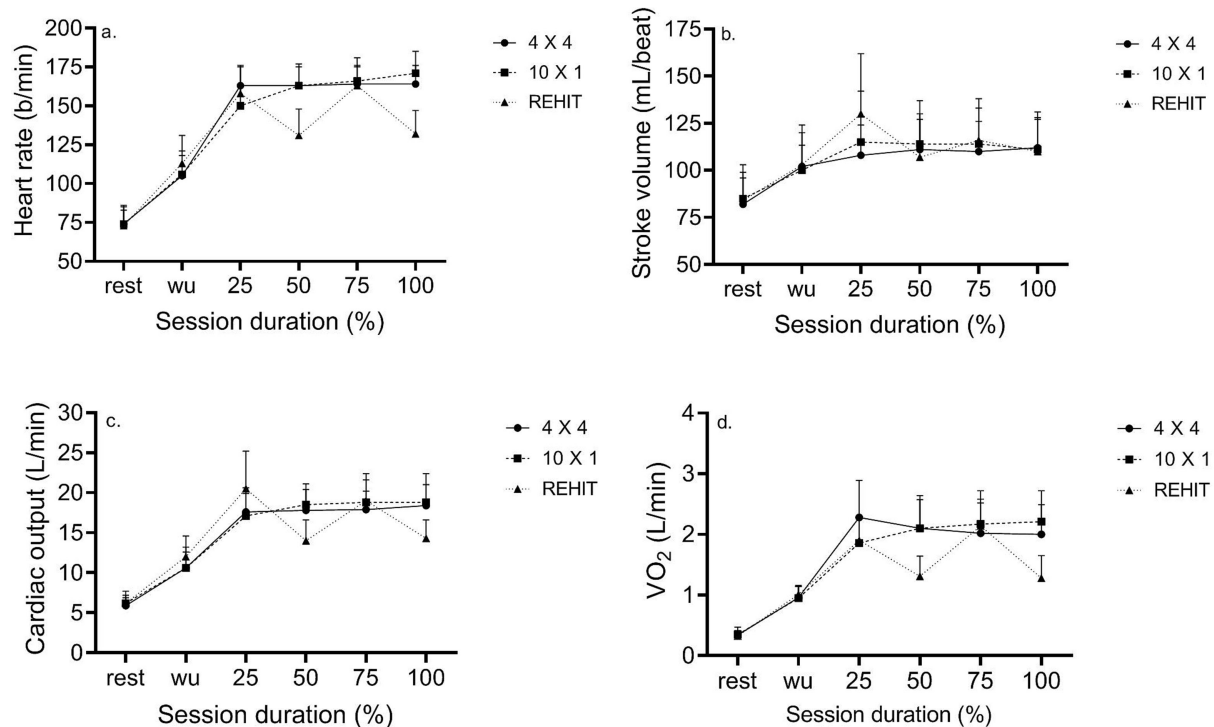


FIGURE 1 Aggregate changes (mean $\pm$ SD) in (A) HR, (B) SV, (C) CO, and (D) VO₂ in 15 men and 13 women undergoing different HIIE protocols. WU, warm-up; These data were combined across men and women due to no significant effect (*p* > 0.05) of sex on these outcomes.

(*p* = 0.097) or sex (*p* = 0.277) and there was no protocol X sex interaction (*p* = 0.277). Peak absolute SV was different across protocols as REHIT revealed higher values than 4 $\times$ 4 (*p* = 0.004, *d* = 0.66), and men exhibited significantly higher values than women (*p* = 0.002), yet no protocol X sex interaction was found (*p* = 0.559). Relative SV (%SV_{max}) was significantly different across protocols (REHIT > 4 $\times$ 4; *p* = 0.005, *d* = 0.78), but there was no main effect of sex (*p* = 0.116) or protocol X sex interaction (*p* = 0.652). Peak HR (b/min) showed a protocol effect (REHIT < 10 $\times$ 1, *p* < 0.001, *d* = 0.59), yet no effect of sex (*p* = 0.197) or protocol X sex interaction (*p* = 0.427). Results showed that relative HR (%HR_{max}) had a main effect of protocol (REHIT < 10 $\times$ 1; *p* < 0.001, *d* = 1.1), yet no main effect of sex (*p* = 0.526) or protocol X sex interaction

(*p* = 0.483). Peak a-vO₂diff differed across protocols (main effect, *p* = 0.011) as REHIT elicited lower values than both 4 $\times$ 4 and 10 $\times$ 1 (*p* = 0.011 and 0.016, respectively). There was no protocol X sex interaction (*p* = 0.332), yet there was a main effect of sex (*p* = 0.011) as men revealed higher values than women. Peak CI showed no main effect of protocol (*p* = 0.101) or sex (*p* = 0.522), and there was no protocol X sex interaction (*p* = 0.976).

3.3 | Mean hemodynamic and metabolic data

There was a main effect of protocol for mean absolute VO₂ (REHIT < 10 $\times$ 1 and 4 $\times$ 4; *p* < 0.001 and *p* < 0.001; *d* = 0.83

TABLE 4 Peak and mean hemodynamic and metabolic responses to various HIIE protocols (mean $\pm$ SD).

	Variable	Men (<i>n</i> = 15)			Women (<i>n</i> = 13)		
		4 $\times$ 4	10 $\times$ 1	REHIT	4 $\times$ 4	10 $\times$ 1	REHIT
Peak	VO ₂ (L/min)*	2.7 $\pm$ 0.5	2.6 $\pm$ 0.5	2.4 $\pm$ 0.4	1.9 $\pm$ 0.3	1.9 $\pm$ 0.2	1.8 $\pm$ 0.2
	VO ₂ (%max)	86 $\pm$ 10	84 $\pm$ 13	81 $\pm$ 11	81 $\pm$ 9	82 $\pm$ 10	80 $\pm$ 10
	CO (L/min)*	19.7 $\pm$ 2.5	21.3 $\pm$ 3.7	21.7 $\pm$ 3.2	17.7 $\pm$ 2.3	17.9 $\pm$ 2.4	18.5 $\pm$ 3.2
	CO (%max)	95 $\pm$ 10	103 $\pm$ 14	105 $\pm$ 18	94 $\pm$ 11	95 $\pm$ 12	98 $\pm$ 16
	SV (mL/beat)*	123.1 $\pm$ 10.9**	132.2 $\pm$ 28.5	139.1 $\pm$ 15.0	105.7 $\pm$ 17.6**	107.4 $\pm$ 17.9	113.2 $\pm$ 17.5
	SV (%max)	102 $\pm$ 12**	109 $\pm$ 19	115 $\pm$ 16	99 $\pm$ 10**	100 $\pm$ 7	106 $\pm$ 14
	HR (b/min)	162 $\pm$ 12	168 $\pm$ 18**	160 $\pm$ 14	171 $\pm$ 10	174 $\pm$ 9**	165 $\pm$ 11
	HR (%max)	91 $\pm$ 3	94 $\pm$ 4**	90 $\pm$ 3	93 $\pm$ 2	95 $\pm$ 4**	90 $\pm$ 4
	a-vO ₂ diff (mL/dL)	14.8 $\pm$ 4.3**	13.8 $\pm$ 1.9**	12.2 $\pm$ 2.1	11.6 $\pm$ 1.9**	11.3 $\pm$ 1.7**	10.7 $\pm$ 1.6
	CI (L/min/m ²)	10.1 $\pm$ 1.3	11.0 $\pm$ 1.9	11.2 $\pm$ 1.7	10.6 $\pm$ 1.2	10.7 $\pm$ 1.5	11.0 $\pm$ 1.6
Mean	VO ₂ (L/min)*	2.1 $\pm$ 0.3**	2.2 $\pm$ 0.4**	1.8 $\pm$ 0.3	1.5 $\pm$ 0.2**	1.6 $\pm$ 0.2**	1.3 $\pm$ 0.1
	VO ₂ (%max)	68 $\pm$ 6**	70 $\pm$ 8**	59 $\pm$ 7	63 $\pm$ 5**	68 $\pm$ 7**	60 $\pm$ 7
	CO (L/min)*	17.6 $\pm$ 2.0	18.1 $\pm$ 2.7**	17.1 $\pm$ 2.7	15.6 $\pm$ 2.0	15.7 $\pm$ 2.3**	14.8 $\pm$ 1.8
	CO (%max)	83 $\pm$ 10	84 $\pm$ 8**	80 $\pm$ 8	84 $\pm$ 9	84 $\pm$ 7**	80 $\pm$ 9
	SV (mL/beat)*	118.0 $\pm$ 10.6	120.9 $\pm$ 16.5	123.3 $\pm$ 18.3	100.1 $\pm$ 16.8	98.7 $\pm$ 17.6	103.0 $\pm$ 15.6
	SV (%max)	97 $\pm$ 10	99 $\pm$ 11	101 $\pm$ 14	95 $\pm$ 11	93 $\pm$ 5	98 $\pm$ 9
	HR (b/min)	149 $\pm$ 12**	149 $\pm$ 14**	138 $\pm$ 13	157 $\pm$ 11**	160 $\pm$ 11**	145 $\pm$ 13
	HR (%max)	84 $\pm$ 1**	84 $\pm$ 3**	78 $\pm$ 5	85 $\pm$ 2**	87 $\pm$ 4**	78 $\pm$ 6
	a-vO ₂ diff (mL/dL)	12.0 $\pm$ 2.6	12.1 $\pm$ 1.7**	10.6 $\pm$ 1.9	9.4 $\pm$ 1.4	10.0 $\pm$ 1.5**	9.1 $\pm$ 1.1
	CI (L/min/m ²)	9.0 $\pm$ 1.0	9.2 $\pm$ 1.3**	8.7 $\pm$ 1.1	9.4 $\pm$ 1.0	9.4 $\pm$ 1.3**	8.9 $\pm$ 0.9

p* < 0.05 versus women.; *p* < 0.05 versus REHIT.

and 0.57) and a main effect of sex (*p* < 0.001), yet no protocol X sex interaction (*p* = 0.164) (Table 4). Additionally, relative VO₂ during REHIT was lower than 4 $\times$ 4 and 10 $\times$ 1 (*p* < 0.001 and < 0.001; *d* = 1.05 and 1.4, respectively), yet no effect of sex (*p* = 0.261) or protocol X sex interaction was found (*p* = 0.368). Mean CO (L/min) showed a protocol effect (REHIT < 10 $\times$ 1; *p* = 0.008, *d* = 0.35) and a significant main effect of sex, as men revealed greater values than women (*p* = 0.007), yet there was no protocol X sex interaction (*p* = 0.863). Mean relative CO (%CO_{max}) also showed a main effect of protocol (REHIT < 10 $\times$ 1; *p* = 0.012, *d* = 0.53), but no effect of sex (*p* = 0.899) or protocol X sex interaction (*p* = 0.918). Stroke volume (mL/beat) was not different across protocols (*p* = 0.252) and there was no protocol X sex interaction (*p* = 0.664), although values were greater in men than women (*p* < 0.001). Relative SV (%SV_{max}) showed no effect of protocol (*p* = 0.207), protocol X sex interaction (*p* = 0.668), or main effect of sex (*p* = 0.309). Mean HR (b/min) showed a protocol effect (REHIT < 10 $\times$ 1 and 4 $\times$ 4; *p* < 0.001 and < 0.001; *d* = 0.99 and 0.91, respectively) yet no effect of sex (*p* = 0.065) or protocol X sex interaction (*p* = 0.358). Mean relative HR (%HR_{max}) exhibited a main effect of protocol (REHIT < 10 $\times$ 1 and 4 $\times$ 4; *p* < 0.001 and < 0.001; *d* = 1.59 and 1.58), yet no effect of sex (*p* = 0.195). Mean a-vO₂diff was

TABLE 5 Peak metabolic and cardiovascular values from HIIE protocols (mean $\pm$ SD).

Variable	4 $\times$ 4	10 $\times$ 1	REHIT
BLa (mM)	7.1 $\pm$ 2.3*	8.7 $\pm$ 2.2*	10.6 $\pm$ 2.7*
HR (b/min)	166 $\pm$ 12	170 $\pm$ 15*	163 $\pm$ 12
HR (%max)	92 $\pm$ 3	94 $\pm$ 4	90 $\pm$ 4**

p* < 0.05 versus other protocols.; *p* < 0.05 versus 10 $\times$ 1.

different across protocols (*p* = 0.01), as REHIT revealed lower values than 4 $\times$ 4 (*p* = 0.003), but there was no protocol X sex interaction (*p* = 0.343) despite a main effect of sex (*p* < 0.001). Mean CI differed across protocol, as REHIT elicited lower values than 10 $\times$ 1 (*p* = 0.006), but there was no main effect of sex (*p* = 0.682) or protocol X sex interaction (*p* = 0.869).

3.4 | Peak metabolic and cardiovascular data

Table 5 reveals that REHIT elicits higher (*p* < 0.001) BLa versus 10 $\times$ 1 (*d* = 0.79) and 4 $\times$ 4 (*d* = 1.42), yet lower (*p* < 0.001) relative HR versus the other protocols (*d* = 1.13 and 0.63, respectively). Peak HR was significantly higher

($p < 0.001$) in response to 10×1 versus all other protocols. No main effect of sex was shown for BLA ($p = 0.05$), absolute HR ($p = 0.19$), or relative HR ($p = 0.52$). Peak PO from REHIT was equal to 8.0 ± 1.6 W/kg versus 6.9 ± 0.6 W/kg; ($p = 0.012$; $d = 0.92$) in men and women, respectively, with values similar to prior studies.³⁷

4 | DISCUSSION

The primary purpose of this study was to examine potential sex differences in the acute CO response to HIIE, which may shed light on previously reported differences in the VO_{2max} response to MICT or HIIT between men and women.^{18,19} The secondary purpose was to identify potential differences in acute hemodynamic response between different HIIE paradigms. Our results show that SV and CO expressed as percentages of maximal values do not differ in men versus women, suggesting a similar hemodynamic response to vigorous exercise.

4.1 | Sex differences in hemodynamic responses to HIIE protocols

Prior findings suggest that HIIE protocols are not interchangeable in men versus women due to inherent differences in cardiovascular and metabolic responses to exercise, which may diminish the magnitude of physiological stress in women compared to men.³⁸ This may be due to women's enhanced resynthesis of ATP between bouts compared to men independent of intensity and perceived exertion.³⁹ Our data oppose our hypothesis yet corroborate prior results¹ showing a higher absolute SV and CO in men in response to three unique HIIE protocols. However, these differences disappeared when outcomes are expressed according to %SV/COMax. In addition, no sex differences were exhibited in peak % VO_{2max} across protocols or peak/mean CI. In the present study, relative VO_{2max} was similar between sexes which may minimize the resultant magnitude of differences in the acute response to HIIE. In addition, there is potential for a ceiling effect leading to no sex differences in these outcomes. In untrained adults, SV typically plateaus at 50% VO_{2max} ,⁴⁰ and the interval intensities are well above this, leading to similar SV responses to exercise. Moreover, SV attained near-maximal values by the first or second bout of 10×1 and 4×4 . Overall, our data suggest that reported sex differences^{18,19} in the VO_{2max} response to HIIE may not be due to unique acute hemodynamic responses between men and women.

One explanation for this finding is the significantly higher peak and mean power output revealed by men

compared to women in response to REHIT.⁴¹ It is likely that sex differences in muscle fiber type, oxidative capacity, and muscle mass explain this result.³⁶ Lactate has also been identified as an energy sensor associated with molecular signaling⁴² which is related to the adaptive response and may partially explain why a prior study¹⁹ examining SIT exhibited significantly greater increases in VO_{2max} in men versus women. These BLA data suggest that women experience less metabolic perturbation than men during interval protocols requiring intensities greater than that associated with VO_{2max} , which is the case with all forms of SIT including REHIT. However, HIIE protocols requiring longer bouts at lower absolute intensities, such as 4×4 ,³⁴ may not require any modification due to a similar BLA response between men and women. Further work is needed using muscle biopsy to elucidate metabolic responses to HIIT between men and women.

4.2 | Differences in hemodynamic responses to various HIIE protocols

Our data show that REHIT elicits a higher relative peak CO and SV versus 4×4 , but a lower mean relative VO_2 (Table 4). These data refute our hypothesis that SIE would elicit lower COMax than higher-volume HIIE. Adami et al.⁴³ required active men to complete cycling at 120% PPO, and data revealed a faster time constant for CO than that of VO_2 kinetics. In active adults, Astorino et al.³¹ reported significantly higher SV and lower VO_2 in response to the Wingate test versus values derived from VO_{2max} testing. This lower VO_2 is also attributed to lower reliance on aerobic metabolism during REHIT, as this bout elicits intensities above 200% PPO. Additionally, our data show that REHIT elicits lower peak a- VO_{2diff} compared to 10×1 and 4×4 , and a lower mean a- VO_{2diff} versus 10×1 . Zafeiridis et al.²⁴ reported similar peak/mean a- VO_{2diff} between HIIE bouts at 95% and 110 %Wmax, although these were matched for effort which was not done in the present study. Falz et al.²³ reported mean and peak a- VO_{2diff} from 4×4 equal to 11.1 and 13.3 mL/dL, similar to our values (Table 4). Data from Bostad et al.¹⁹ showed that both men and women reveal a significant and similar increase in a- VO_{2diff} after 6 and 12 weeks of SIT. In addition, there was a positive correlation between the change in peak a- VO_{2diff} and change in peak VO_2 , likely related to significant increases in mitochondrial content and capillary density demonstrated with chronic interval training.^{17,19,32,44} Overall, the circulatory system rapidly adjusts to these high intensities to optimize oxygen delivery to the exercising legs, which is coincident with increases in O_2 extraction, allowing attainment of maximal CO values (Table 4).

The attainment of peak SV and CO during acute interval exercise is significant because peak cardiovascular stress may be important for optimizing changes in $\text{VO}_{2\text{max}}$ in response to exercise training.²⁵ We identified peak values as the highest of the average of three consecutive values at either 25%, 50%, 75%, or 100% of protocol duration. Previous studies^{23,24} frequently report peak values of SV and CO, yet do not denote how the values are identified. This may cause uncertainty related to how the outcome is calculated and result in potential discrepancies in values across studies. Additional discrepancies may be due to differences in exercise protocol, modality, or participants' sex, age, and fitness level across studies.⁴⁵ Similarly, prior research^{24,46} reports mean SV and CO, but does not denote whether these values represent the average during work and recovery throughout the entire session.

Reduced exertion high-intensity training exhibited a significantly lower mean relative HR, VO_2 , CO, and CI versus the HIIE protocols. This may be due to the relatively brief vigorous efforts attendant with REHIT (40s) that are accompanied by 9 min and 20s of pedaling at 20% PPO, eliciting a relatively low average cardiometabolic stimulus versus 10×1 and 4×4. Despite differences in interval duration (10 vs. 16 min) and intensity (85 vs. ~60% PPO), 10×1 and 4×4 reveal similar peak and mean hemodynamic and metabolic responses (Table 4). However, REHIT demonstrates similar peak relative VO_2 , absolute/relative SV, and absolute/relative CO versus 10×1, with significantly lower values seen in 4×4 (Table 4). Across protocols, 4×4 had the lowest overall relative power output which may mediate the lower acute hemodynamic response shown versus 10×1 and REHIT. Zafeiridis et al.²⁴ required highly active men ($\text{VO}_{2\text{max}} = 49 \text{ mL/kg/min}$) to complete repeated 30s efforts followed by 30s recovery at 110% PPO and 2 min efforts at 95% PPO followed by 2 min of recovery, which were matched for effort. Peak and average HR were similar across protocols, yet the longer intervals elicited higher VO_2 , mean SV, and mean/peak CO compared to the shorter efforts, supporting prior data.⁴⁷ It is possible that the high peak VO_2 and SV/CO characteristic of REHIT, combined with high BLA, are partially responsible for significant increases in $\text{VO}_{2\text{max}}$ observed long-term. However, its lower mean HR, SV, and CO compared to higher volume HIIE protocols induce less overload on the cardiovascular system, thereby attenuating increases in $\text{VO}_{2\text{max}}$ than 4×4,⁴⁸ suggesting that the training-induced increase in $\text{VO}_{2\text{max}}$ may be associated more with the average magnitude of stress on the cardiovascular system rather than absolute power output.

Table 4 shows peak SV exceeding values acquired from $\text{VO}_{2\text{max}}$ testing as well as peak CO values approaching or equivalent to maximal values. Falz et al.²³ reported 10% higher peak SV/CO in response to the 4×4 protocol than

values derived from incremental exercise, supporting our findings. In both studies, thoracic impedance was used to estimate hemodynamic responses. Fontana et al.⁴⁹ reported significantly higher SV (+30 mL, ~35% higher) and similar CO in response to a single Wingate test compared to that from incremental exercise. They used Innocor (inert gas rebreathing) which like thoracic impedance, provides an indirect and noninvasive way to estimate SV and CO. Additional research in highly-trained men demonstrated similar near-maximal peak CO and SV values in response to acute HIIE and SIE; however, they used arterial pressure wave analysis to estimate SV and CO.³¹ Faisal et al.⁵⁰ revealed an immediate overshoot in SV with initiation of exercise at 85 % $\text{VO}_{2\text{max}}$ due to activation of the muscle pump, which leads to a rapid acceleration in CO. Subsequently, SV declines and then plateaus. It is plausible that the “all-out” 6s sprint preceding each bout of REHIT followed by application of the load elicits a similar overshoot in SV, leading to values above that derived from ramp testing. It is apparent that a single assessment of $\text{VO}_{2\text{max}}$ may underestimate the true value due to a learning effect,⁵¹ and consequently, our initial measure of $\text{VO}_{2\text{max}}$ and SV/CO may not elicit true ‘maximal’ values for some participants. Our results also show that mean SV obtained from HIIE protocols differing in training volume, recovery duration, and intensity ranged from 93% to 101% of maximal values, suggesting a large aggregate strain on the cardiovascular system throughout each protocol. Previous research²⁵ documents that the greatest gains in $\text{VO}_{2\text{max}}$ arise from exercise done at high percentages of $\text{VO}_{2\text{max}}$ due to substantial stress on the oxygen transport system. Our research suggests that these adaptations may be more dependent on a high SV or peak VO_2 response, since average VO_2 ranged from 59% to 70% $\text{VO}_{2\text{max}}$.

4.3 | Limitations

The study has a few limitations. First, data were acquired in healthy, active adults, so results cannot be applied to other populations. Second, despite these protocols being widely applied in sports science, there are infinite permutations of interval exercise that may elicit discrepant hemodynamic responses. These protocols also have different intensities, interval and recovery durations, as well as exercise volume which may make comparisons across protocols challenging. Third, men typically have a higher $\text{VO}_{2\text{max}}$ than women due to a higher absolute SV and CO,⁸ so different results could occur if men had a higher $\text{VO}_{2\text{max}}$ than women. In addition, treadmill exercise elicits true maximal CO and VO_2 versus cycling, so our data only apply to this mode of exercise. Fourth, our men and women had a significant difference in age, although all are considered young and healthy. However, adults ranging from 20- to 80-year-old showed no

age-related difference in the $\text{VO}_{2\text{max}}$ response to 8 weeks of HIIT,⁵² indicating that age may not impact the adaptability of the cardiovascular system to HIIT. We also did not consider the effects of the menstrual cycle or oral contraceptives, and 10 of 13 women were eumenorrheic. However, data⁵³ show no effect of menstrual phase on time to exhaustion during graded exercise or maximal SV, CO, and VO_2 ,⁵⁴ suggesting minimal effects on our outcomes. Lastly, no method of acquiring hemodynamic responses including thoracic impedance has been validated during brief repeated intervals such as REHIT, in which there is no steady-state, rapid attainment of supramaximal power outputs and near-maximal leg blood flow, and dramatic lag in the VO_2 and CO response. Mean values are derived from the entire 10 min protocol, but 'peak' values are obtained from only three data points, so the true accuracy of peak SV/CO obtained from sprint interval exercise which may exceed 'maximal' values remains to be determined. Despite these limitations, this study is strengthened by precise determination of intensities for the exercise protocols, implementation of three widely-used interval protocols, and careful execution of methods used to estimate CO from thoracic impedance.

5 | CONCLUSION

Our data show no significant differences in relative SV, CO, or VO_2 across interval protocols between men and women, suggesting a similar hemodynamic response. In addition, results show that SIE eliciting power outputs approximately 2-fold above that associated with $\text{VO}_{2\text{max}}$ reveals the highest CO values yet lowest VO_2 , likely due to greater reliance on nonoxidative metabolism. Previously reported differences^{18,19} in the $\text{VO}_{2\text{max}}$ response to interval exercise may not be due to a unique hemodynamic or VO_2 response between men and women.

6 | PERSPECTIVES

High-intensity interval training has become popular for its ability to increase endurance performance and $\text{VO}_{2\text{max}}$ while frequently requiring less time compared to MICT. Research is equivocal; however, whether sex differences exist in the $\text{VO}_{2\text{max}}$ response to HIIT.⁴⁰ Previous research recommended that HIIT protocols be individualized based on sex in order to elicit similar responses.³⁸ Our data suggest that this may not be the case for higher-volume HIIE protocols due to similar blood lactate and hemodynamic responses. Nevertheless, when HIIE protocols require short bursts at intensities above $\text{VO}_{2\text{max}}$, such as REHIT, power output and BLa concentration are different between sexes, which may necessitate that women undergo higher relative

intensities to equalize the metabolic stress. This study will inform development of interval exercise prescription to achieve the greatest health and fitness benefits for adults.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

All data related to this study are contained in the manuscript. The data that support the findings of this study are available from the corresponding author upon reasonable request.

INFORMED CONSENT

Participants provided written informed consent before initiating the study.

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